

# Distributed Control Framework for Connectivity Preservation and Collective Reconfiguration in Flow-Driven Robot Fleets

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**Abstract**—We study the deployment of underwater robot fleets in large, flow dominated environments where localization is denied, communication is limited, and energy is scarce. The robots need to travel long distances, detect emerging targets, and stay connected using only local information. To meet these challenges we design a distributed framework that exploits natural drift and activates control only when required. The method begins with the robots deployed as a fully connected network. Using minimal information exchange, each robot prunes non-critical edges, producing a sparse and tree like structure that spans the entire team. Upon this structure we construct a distributed control Lyapunov function–control barrier function (CLF–CBF) framework. Each robot enforces barrier constraints to maintain the necessary connections, while redundant links are discarded when alternative paths emerge. When a target is discovered, this framework ensures the detecting robot attends the target, while the team reconfigures to provide a connected formation. This approach guarantees long term connectivity with very low control effort, and enables collective response to dynamic tasks. Simulations validate the framework and demonstrates that it yields spanning structures that converge toward a minimum spanning tree over time, highlighting both its efficiency and robustness under realistic underwater conditions.

## I. INTRODUCTION

Deploying fleets of underwater robots in deep-ocean and under-ice environments—where GPS is unavailable, communication is intermittent and low-bandwidth, and dynamic flows dominate—poses unique challenges [1]–[3]. Robots must traverse long distances on limited energy, discover regions of interest, and relay data using only local exchanges.

The key difficulties are to (i) minimize control effort by drifting with the flow when possible, (ii) maintain coverage and connectivity over large domains, and (iii) ensure global connectivity using only neighbor-to-neighbor information. Our method lets robots drift collectively, intervening only to stay within a region or preserve links [4]. Prior work demonstrates minimal-control drift in gyres [5], [6] and models dispersion as Brownian diffusion [7]–[11].

Maintaining connectivity and enabling target-driven reconfiguration under energy and communication constraints has leveraged graph-theoretic tools [12], [13] and control barrier functions (CBFs) for connectivity [14]–[16]. Hybrid MST-based CBF frameworks exist [17] but require

global information. Exact distributed MST protocols remain communication-intensive [18]–[22], prompting approximate methods [23].

We exploit two facts: an initially connected deployment and the goal of reducing long-range links. Using only neighbor index exchanges, robots first identify critical edges [24], prune non-essential links into a sparse tree, then run a distributed CLF–CBF controller on this backbone. Upon target detection, the discovering robot advances while the team reconfigures to preserve connectivity and coverage. Simulations show this structure converges to an MST, validating its efficiency and robustness in flow-dominated settings.

This paper is organized as follows. Sec. II formulates the problem. Sec. III details our framework. Sec. IV presents simulations.

## II. BACKGROUND AND PROBLEM FORMULATION

### A. Flow and Dispersion [7]–[9]

For  $N$  robots with positions  $x_i(t) \in \mathbb{R}^n$ , let  $\bar{x} = \frac{1}{N} \sum_i x_i$ . Then

$$\dot{\bar{x}} \approx f_{\text{flow}}(\bar{x}, t), \quad dx_i = f_{\text{flow}}(\bar{x}, t) dt + dW_i,$$

where  $dW_i$  is Brownian noise. In 2D:

$$f_{\text{flow}} = v_{\text{bg}} + \sum_{q=1}^Q \kappa_q \begin{bmatrix} -(x_2 - x_{2,q}) \\ x_1 - x_{1,q} \end{bmatrix},$$

and

$$\dot{x}_i = f_{\text{flow}}(\bar{x}, t) + u_i + d_i, \quad \|d_i\| \leq \bar{d}.$$

### B. Network Model

Robots  $i \in \mathcal{V}$  have  $\|u_i\| \leq u_{\text{max}}$  and maintain separation  $d_{\text{min}}$ . Communication uses a disk of radius  $R_{\text{max}}$ :  $(i, j) \in \mathcal{E}$  if  $\|x_i - x_j\| \leq R_{\text{max}}$ . Neighborhood  $\mathcal{N}(i) = \{j : \|x_i - x_j\| \leq R_{\text{max}}\}$ .  $\mathcal{G}_0$  is connected.

### C. Objectives

Agents know only  $x_i$ ,  $\{x_j - x_i\}_{j \in \mathcal{N}(i)}$ , and local graph data. No global information.

Design a decentralized control that:

- 1) Keeps  $\mathcal{G}_t$  connected.
- 2) Uses flow drift, activating  $u_i$  only as needed.
- 3) Spreads for exploration and reconfigures toward targets while preserving connectivity.

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**Algorithm 1** Distance-First Pruning

- 1: Identify critical set  $\mathcal{C}$  [24]
- 2: Rank  $\mathcal{E}^-$  by length; build vertex-disjoint batch
- 3: Prune edges satisfying all four conditions
- 4: Recoup isolated nodes; local rewiring with score  $S_{\text{add}} = \alpha S_{\text{distance}} + \beta S_{\text{connectivity}} + \gamma S_{\text{efficiency}}$

**Algorithm 2** Decentralized Control Framework

- 1: Update neighbors; identify critical edges
- 2: Prune with guards; set  $\mathcal{C}_i$  from pruned graph
- 3: Solve CLF-CBF QP: CLF if target assigned, CBFs for safety/connectivity
- 4: Apply  $u_i$ ; broadcast state

## III. METHODOLOGY

Our method has two components: sparse critical structure identification via edge pruning, then distributed CLF-CBF control on this structure.

## A. Distributed Critical Structure Identification

We identify critical edges whose removal disconnects the network using [24]. For non-critical edges  $\mathcal{E}^- = \mathcal{E} \setminus \mathcal{C}$ , we prune based on distance with guards.

Edge  $(i, j) \in \mathcal{E}^-$  is removed if:

- 1) **Alternative:** Path  $i \rightsquigarrow j$  exists with  $\leq H$  hops where  $\max_{(p,q) \in \text{path}} \|x_p - x_q\| < \|x_i - x_j\|$ .
- 2) **Degree:**  $\min\{\deg(i), \deg(j)\} \geq 2$ .
- 3) **Margin:** Alternative edges satisfy  $\|x_p - x_q\| \leq R_{\text{max}} - \varepsilon$ .
- 4) **Non-critical:**  $(i, j) \notin \mathcal{C}$ .

Batching ensures no two pruned edges share vertices. Optional triangle preservation uses score  $P(i, j) = \frac{|\mathcal{N}_i \cap \mathcal{N}_j|}{\min\{\deg(i), \deg(j)\}} (1 + |\mathcal{N}_i \cap \mathcal{N}_j|)$  as tie-breaker.

## B. CLF-CBF Control on Sparse Structure

On pruned graph  $\mathcal{G}'$ , robot  $i$  maintains critical neighbors  $\mathcal{C}_i = \{j : (i, j) \in \mathcal{E}'\}$  and sensing neighbors  $\mathcal{N}_i$ .

Barriers: Safety  $h_{ij}^{\text{safe}} = \|x_i - x_j\|^2 - d_{\min}^2$  for  $j \in \mathcal{N}_i$ , connectivity  $h_{ij}^{\text{conn}} = R_{\text{max}}^2 - \|x_i - x_j\|^2$  for  $j \in \mathcal{C}_i$ .

Lyapunov:  $V_i = \|x_i - x_i^{\text{ref}}\|^2$  where  $x_i^{\text{ref}} = x_i$  (drift) or  $x_i^{\text{goal}}$  (target).

**Exploration:**  $V_i \equiv 0$ , robots drift with CBF constraints.

**Target:** Activate CLF toward goal while preserving connectivity.

## IV. SIMULATION SETUP AND RESULTS

We simulate  $N = 12$  (or 3–20 random) robots in a  $10 \times 10$  m workspace with  $\Delta t = 0.1$  s. Hydrodynamic parameters (mass  $2.0 \pm 0.2$  kg, drag  $C_d = 0.8 \pm 0.1$ , area  $0.01 \pm 0.002$  m<sup>2</sup>) and speed cap 2 m/s model realistic behavior. Communication uses range  $R_{\text{max}} = 1.2$  m with attenuation, ensuring at least  $0.3R_{\text{max}}$ . Safety distance is  $d_{\min} = 0.6$  m.

Targets invoke a CLF-CBF QP ( $\alpha = 1.5, \beta = 5.0, \beta_s = 10.0, \|u\| \leq 0.40$ , cost  $\|u\|^2 + 50\gamma^2$ ) solved via CVXPY.

Exploration uses CBF-based force control. The flow field combines a base current, two Gaussian vortices, and sinusoidal turbulence.

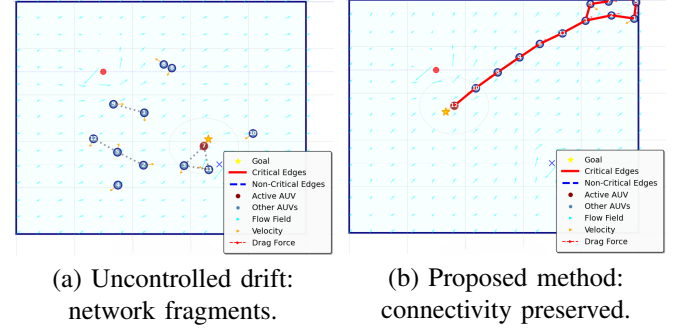


Fig. 1. Comparison of uncontrolled drift vs. decentralized CLF-CBF.

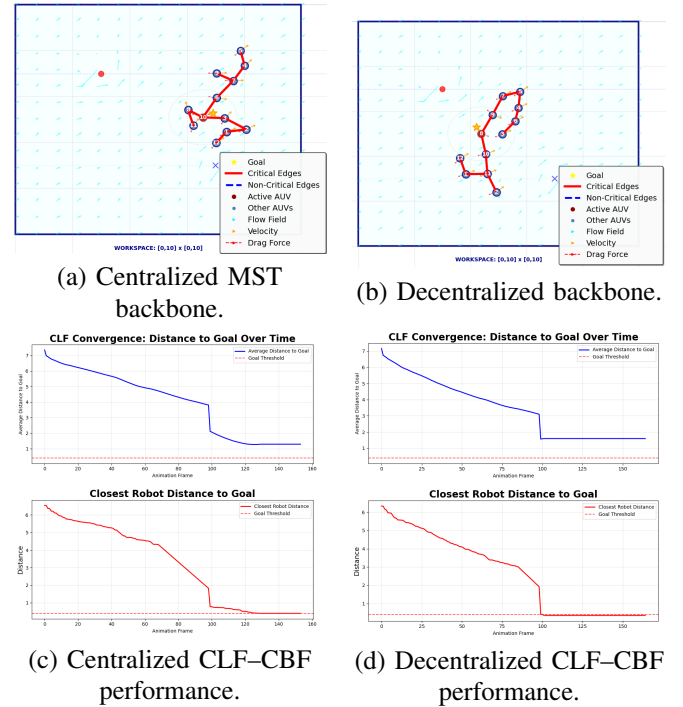


Fig. 2. Centralized vs. decentralized backbones: (a,b) final networks; (c,d) robot-target distances.

In all cases, our distributed framework preserves connectivity and achieves performance comparable to centralized baselines.

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